# Final exercise

## Concepts of Biostatistics and Data Science: a two-day crash course in R

> Data: the simulated Healthy Hearts District Study (teaching only). Template script: R/08_final_exercise.R. Use the cheat sheet and decision diagrams.

## Final exercise: choose, plot, conclude (25 minutes)

Work alone or in pairs. Pick ONE question. Use the cheat sheet and decision diagrams, and the template script `R/08_final_exercise.R`. There is more than one defensible analysis for each question; what matters is that you can justify yours.

| | Question |
|---|---|
| F1 | Were participants in the lifestyle programme more satisfied with their care at 12 months than those given standard advice? |
| F2 | Do men and women differ in C-reactive protein (CRP) at baseline? |
| F3 | Is a healthier diet score associated with lower baseline systolic blood pressure, after allowing for age and sex? |
| F4 | Was high blood pressure at 12 months less common in the lifestyle programme group than in the standard advice group? |

Step 1. Write your question and the outcome, its type, the groups or exposure, and the design (independent or paired; any confounders).

Your answer

Step 2. Which analysis did you choose, and why? Name one alternative and say why you did not choose it.

Your answer

Step 3. Describe the data: give the numbers you will report (with denominators or n).

Your answer

Step 4. Produce one plot that shows the answer. Paste it here or sketch it.

Plot

Step 5. Results: the effect estimate with its 95% CI (and p-value if you wish). What did you check?

Your answer

Step 6. Conclusion in plain language (3 to 5 sentences) for a reader with no statistical training. Include the size of the effect, the uncertainty, whether it matters in practice, and one limitation.

Conclusion